Let $f : \mathbb{R} \to \mathbb{R}$ where $f(x) = x + \frac{1}{x}$. Compute:

$$\int_0^1 \frac{f(f(x))}{f(x)} dx$$

$$\frac{f(f(x))}{f(x)} = \frac{x + \frac{1}{x} + \frac{1}{x + \frac{1}{x}}}{x + \frac{1}{x}}$$

$$\frac{f(f(x))}{f(x)} = 1 + \frac{\frac{1}{x + \frac{1}{x}}}{x + \frac{1}{x}}$$

$$\frac{f(f(x))}{f(x)} = 1 + \frac{1}{x^2 + 2 + \frac{1}{x^2}}$$

$$\frac{f(f(x))}{f(x)} = 1 + \frac{x^2}{x^4 + 2x^2 + 1}$$

$$\frac{f(f(x))}{f(x)} = 1 + \frac{x^2}{x^4 + 2x^2 + 1}$$

$$\frac{f(f(x))}{f(x)} = 1 + \frac{x^2}{\left(x^2 + 1\right)^2}$$

$$\frac{f(f(x))}{f(x)} = 1 + \frac{x^2}{\left(x^2 + 1\right)^2}$$

$$\int_0^1 \left(1 + \frac{x^2}{\left(x^2 + 1\right)^2}\right) dx$$

$$1 + \int_0^1 \frac{x^2}{\left(x^2 + 1\right)^2} dx$$

$$x = \tan\theta \quad dx = \sec^2\theta d\theta$$

$$1 + \int_0^{\frac{\pi}{4}} \frac{\tan^2\theta}{\sec^4\theta} \sec^2\theta d\theta$$

$$1 + \int_0^{\frac{\pi}{4}} \frac{\tan^2\theta}{\sec^2\theta} d\theta$$

$$1 + \int_0^{\frac{\pi}{4}} \sin^2\theta d\theta$$

$$1 + \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2\theta}{2} d\theta$$

$$1 + \frac{\pi}{8} - \frac{1}{2} \int_0^{\frac{\pi}{4}} \cos 2\theta d\theta$$

$$u \coloneqq 2\theta \quad du = 2d\theta$$

$$1 + \frac{\pi}{8} - \frac{1}{4} \int_0^{\frac{\pi}{2}} \cos u du$$

$$1 + \frac{\pi}{8} - \frac{1}{4} \sin u \ |^{\frac{\pi}{2_0}}$$

$$1 + \frac{\pi}{8} - \frac{1}{4}$$

$$\frac{3}{4} + \frac{\pi}{8}$$